[This question paper contains 4 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : **216** **K**

Unique Paper Code : 2932812302

Name of the Paper : Digital System Design

Name of the Course : **B.Tech – CSE (DSC-8)**

Semester : III

Duration : 3 Hours Maximum Marks : 90

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.
2. Attempt **five** questions in all, including question no. **1**, which is compulsory.
3. **All** questions carry equal marks.

1. (a) Draw the general block diagrams of a Mealy and Moore state machine and explain how the output depends on inputs and states in each case, with the help of a suitable example.

   (b) Write a Verilog code to implement a Full Adder. Write a testbench to test the working your code. There should be no syntax errors.

   (c) Using only 2 × 1 MUX design the following gates:
   (i) NAND (ii) NOR (iii) XOR

   (6 × 3 = 18)

2. (a) The following microoperation is to be implemented:
   LOAD R2, [R3]
   Draw the block diagram of a generic datapath and discuss how the control path orchestrates the different sections of the datapath to implement the given operation.

P.T.O.

(b) Design a 4 – bit Ring and Johnson Counter. Show all intermediate steps. Analyze the counters for self – starting and modify the circuit in – case of a lock – out.

$(9 \times 2 = 18)$

3. (a) Design an arithmetic circuit with one selection variable s and two data inputs A and B. When $s = 0$, the circuit performs the addition operation $F = A + B$. When $s = 1$, the circuit performs the increment operation $F = A + 1$.

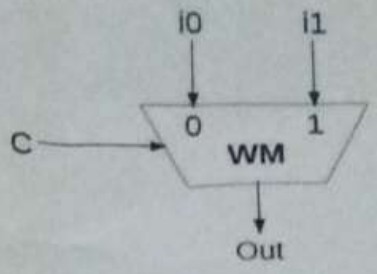
(b) Design a logic circuit with 4 inputs, A, B, C, and D, that will produce output 1 only whenever two adjacent inputs are 1. A and D are also to be treated as adjacent. Implement using universal logic gates.

$(9 \times 2 = 18)$

4. (a) Write any 3 points of difference between the Decoder and Demultiplexer.

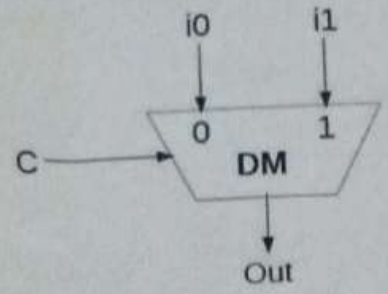
Consider the Boolean Function $F=(a \oplus b).(b \oplus c).(d \oplus e).(e \oplus f)$. You are required to implement the function F by interconnecting 2×1 *MUX* and a 2-to-4 *decoder*. You are given three working 2×1 *MUX* (labeled as WM), four defective 2×1 *MUX* (labeled as DM) and a single defective 2-to-4 *decoder* (labeled as DD), for this purpose. All are shown in the following figure. You are not allowed to use any other gate (not even an inverter) in your implementation. Assume that you have access to $a, a', b, b', c, c', d, d', e, e', f$ and f', so that any literal can be fed directly to the input of *MUX* or the *decoder*.

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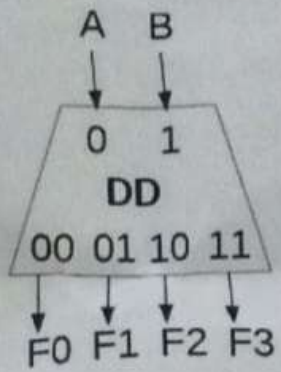
working

Out = (C AND i1) + (NOT(C) AND i0)



Defective

Out = (C AND NOT(i1)) + (NOT(C) AND i0)



F0 = NOT(A) AND NOT(B)
F1 = 0
F2 = 1
F3 = A AND B

Defective.

Discuss how you would connect the (working and/or defective) *MUX* and *decoder* to implement *F* correctly. Label each working/ defective *MUX*, and/or *decoder* correctly as described above.

(b) Using 2's complement arithmetic, solve the following:

(i) Subtract -108_{10} from $+14_{10}$.

(ii) Add -115_{10} with -37_{10}.

$(9 \times 2 = 18)$

5. (a) A sequential circuit has two JK flip-flops, A and B, two inputs x and y, and one output z. The flip-flop input equations and circuit output equations are as follows –

$$J_A = Bx + \bar{B}\bar{y}; K_A = \bar{B}x\bar{y}$$
$$J_B = A\bar{x} \quad ; K_B = A + x\bar{y}$$
$$z = A\bar{x}\bar{y} + B\bar{x}\bar{y}$$

(i) Draw the logic diagram of the circuit.

(ii) Tabulate the state table.

(iii) Derive the state equations for A and B.

(b) Given the Boolean Function:

$$Z = \sum_m(0,1,4,5,6,7,9,11,15) + \sum_d(10,14)$$

(i) Draw the Karnaugh Map

(ii) Determine the minimum sum of product expression for Z.

(iii) Draw the combinational logic circuit for Z using NAND gate.

$(9 \times 2 = 18)$

6 (a) Show any **two** conversions in case of Flip Flops:

(i) J-K to D

(ii) D to J-K

(iii) S-R to J-K

(iv) D to S-R

(b) Write VHDL or Verilog code for a 3:8 Decoder. Develop a testbench that applies all input combinations and displays the outputs. There should be no syntax errors.

$(4.5 \times 4 = 18)$